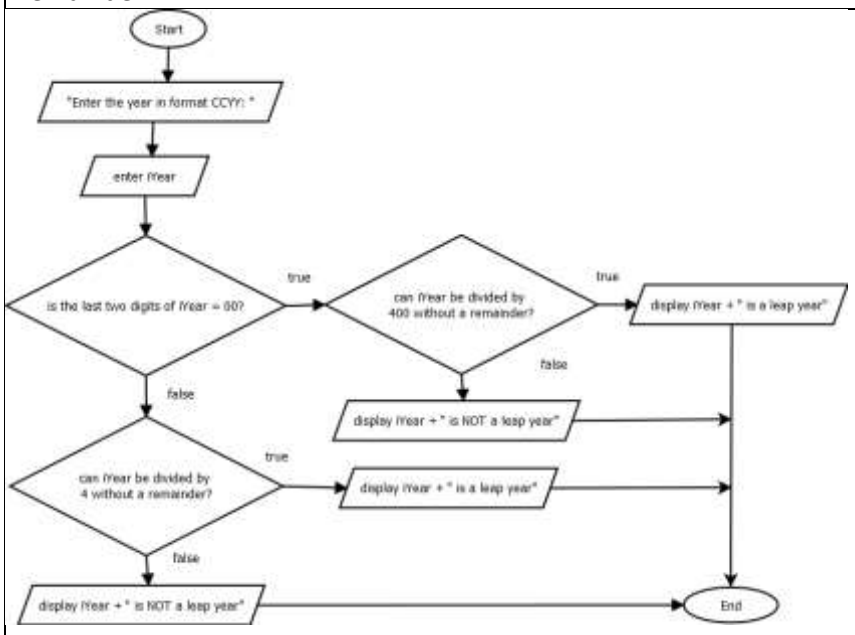
Subject	PPAF05D
Description	40% Summative Assessment opportunity **2** session 1
Date	March 2021
Time	2.5 Hours
Marks ratio	1 mark per 1.5 minutes
Marks	100
Assessors	U Wassermann and GA Ogutu
Internal Moderator	M Liebenberg

Question 1 [10]

A Java program has been distributed to your PC. It is available in the file called **LeapYear.java**.

- Open the program in Notepad++ (do not change any of the code provided)
- Complete the Java program based on the flow chart provided below the code, as well as the comments in the code. NOTE: You must translate the flowchart to Java, do not create your own solution.
- When you are done, copy the code from Notepad++ and paste it into an empty Word document.
- Copy the code from the Word document and paste it into myTUTor in the provided space. *If you do not copy to Word first, all end-of-line characters will be missing in myTUTor.*

This flowchart is an algorithm for a program where the user can enter a year, e.g. 2003, and the program has to determine whether the year is a leap year or not. A leap year is determined as follows: If the year ends in 00 (e.g. 1900 or 2000), it is a leap year if it can be divided by 400 without a remainder. If it does not end in 00, it is a leap year if it can be divided by 4 without a remainder.



Program provided to students

```
import java.util.Scanner;
public class LeapYear
{ //start class
     public static void main(String[] args)
     { //start main method
          Scanner keyboard = new Scanner(System.in);
          int iYear;


     } //end main method
} //end class
```

MEMO and mark sheet	mark	student's mark
`System.out.print("Enter the year: ");`	0.5	
`iYear = keyboard.nextInt();`	1	
`if (iYear % 100 == 0) //year ends in 00`	1	
`{`		
`if (iYear %400 ==0 )`	1	
`System.out.print(iYear + " is a leap year");`	1	
`else`	0.5	
`System.out.print(iYear + " is NOT a leap year");`	1	
`}`		
`else //year does not end in 00`	0.5	
`{`		
`if (iYear % 4 == 0)`	1	
`System.out.print(iYear + " is a leap year");`	1	
`else`	0.5	
`System.out.print(iYear + " is NOT a leap year");`	1	
`}`		
	10	**0**

Question 2 [6]

The following program determines the cost of a number of tickets to a concert. Tickets in block A of the hall costs R150 per ticket. Those in block B are R110 per ticket. A Java program has been distributed to your PC. It is available in the file called **TicketPriceSwitch.java**.

- Open the program in Notepad++. (You can run the program to see the output.)
- Change the switch statement to a nested if..else statement so that the program will still provide the same output.
- Copy **only** the nested if statement to a Word document.
- Copy the code for the nested if statement from the Word document, and then paste it to MyTUTor in the provided space.

If you do not copy to Word first, all end-of-line characters will be missing in myTUTor.

Code provided to students

```
import java.util.Scanner;
import java.text.DecimalFormat;
public class TicketPriceSwitch
{
     public static void main( String[] args)
     {
     Scanner kb = new Scanner(System.in);
     DecimalFormat formatter = new DecimalFormat("R0.00");

     double rTicketPrice;

     System.out.println("How many tickets do you want to buy: ");
     int iNumberTickets = kb.nextInt();
     System.out.println("In which block must the tickets be?: ");
     char cBlock = Character.toUpperCase(kb.next().charAt(0));

     switch (cBlock)
     {
          case 'A': rTicketPrice = 150 * iNumberTickets; break;
          case 'B': rTicketPrice = 110 * iNumberTickets; break;
          default: rTicketPrice = 0; System.out.println("Illegal block
number");
     } //end switch

     System.out.println("You need to pay " +
formatter.format(rTicketPrice));
     } //end main
} //end class
```

MEMO and mark sheet	mark	student's mark
`if (cBlock == 'A')`	1	
`  rTicketPrice = 150 * iNumberTickets;`	0.5	
`else`	0.5	
`  if (cBlock == 'B')`	1	
`  rTicketPrice = 110 * iNumberTickets;`	0.5	
`  else`	0.5	
`  {`	0.5	

rTicketPrice = 0;	0.5		
System.out.println("Illegal block number");	0.5		
} // Braces for 2 statements.	0.5		
	6	**0**	

Question 3 [5]

The following program will ask the user which times table must be displayed, as well as how many values of the table to display. The Java program has been distributed to your PC. It is available in the file called **TimestableWhileLoop.java**. (You can run the program to see the output.)

- Open the program in Notepad++.
- Change the while loop to a for loop so that it will still provide the same output.
- Copy only the for loop to a Word document.
- Copy the code for the for loop from the Word document, and then paste it to MyTUTor in the provided space. *If you do not copy to Word first, all end-of-line characters will be missing in myTUTor.*

Program provided to students

```
import java.util.Scanner;
public class TimestableWhileLoop
{ //start class
     public static void main(String[] args)
     { //start main method
          Scanner keyboard = new Scanner(System.in);

          System.out.print("Which timetable do you want to see? : ");
          int iTimes = keyboard.nextInt();
          System.out.print("How many values of the " + iTimes + " times
table do you want? :");
          int iHowmany = keyboard.nextInt();

          int iAnswer;
          int iCounter = 1 ;

          while (iCounter <= iHowmany)
          {
              iAnswer = iCounter * iTimes;
               System.out.println(iCounter + " times " + iTimes + " = " +
iAnswer);
               iCounter ++;
          }

     } //end main method
} //end class
```

MEMO and mark sheet	mark	student's mark
int iCounter; Declare iCounter (no marks if = 1 kept)	1	
for (iCounter = 1 ;	1	
iCounter <= iHowmany;	1	

iCounter ++)	1	
{		
iAnswer = iCounter * iTimes;		
System.out.println(iCounter + " times " + iTimes + " = " + iAnswer);		
} iCounter ++ removed	1	
	5	**0**

Question 4 [10]

The Java program called **HCFWhileWithErrors.java** that was distributed to your PC, attempts to find and display the common factors of two values. It should also display the highest common factor.

The program has some **syntax** as well as **logical errors**. Your task is to correct all syntax and logical errors. The screenshot alongside shows the output it should produce.

```
Enter first value: 12
Enter second value: 30
A common factor 1 found.
A common factor 2 found.
A common factor 3 found.
A common factor 6 found.

4 common factors were found.
HCF(12,30) = 6
```

- Open the program in Notepad++.
- Correct all the errors so that the program contains no syntax errors, and the correct output is displayed.
- When you are done, copy the code from Notepad++ and paste it into an empty Word document.
- Copy the code from the Word document and paste it into myTUTor in the provided space.

If you do not copy to Word first, all end-of-line characters will be missing in myTUTor.

MEMO and mark sheet	mark	student's mark
import java.util.Scanner;		
public class HCFWhileWithErrors		
{ //start class		
public static void main(String[] args)		
{//start main method		
Scanner keyboard = new Scanner(System.in);		
int iHCF = 0, iHowMany **=** **0** ;	1	
System.out.print("Enter first value: ");		
int iFirstNum = keyboard.nextInt();	1	
System.out.print("Enter second value: ");		
int iSecondNum = keyboard.nextInt();		
int iCount = **1**; //not 3	1	
while (iCount <= Math.**m**in(iFirstNum,iSecondNum)) //not **M**in	1	
{		
if (iFirstNum % iCount **==** 0 && iSecondNum % iCount == 0) **//must be == not =**	1	
{		
iHCF = iCount;		

iHowMany++; //not hwMany	1	
System.out.println("A common factor " + iCount + " found."); //not ,	1	
}		
iCount++ ;		
} //end	1	
System.out.println ("\n" + iHowMany + " common factors were found."); //add ; at end	1	
System.out.println ("HCF(" + iFirstNum + "," + iSecondNum + ") = " + iHCF); //not iHowMany	1	
} //end main method		
} //end class		
	10	0

Question 5 [16]

- Open the file called **HolidayBooking.java** that was distributed to your PC in Notepad++ (do not change any of the code provided)
- Complete the Java program based on the instructions that follow, and the comments in the code. **READ the comments in the code!**
- When you are done, copy the code from Notepad++ and paste it into an empty Word document.
- Copy the code from the Word document and paste it into myTUTor in the provided space. *If you do not copy to Word first, all end-of-line characters will be missing in myTUTor.*

This program should display a menu, and allow a person to repeatedly add a number of adults or a number of children to the booking of a holiday trip.
Here are the detailed instructions:

The program must start by asking how many nights a person wants to sleep on the trip.

```
How many nights do you want to sleep? :5
```

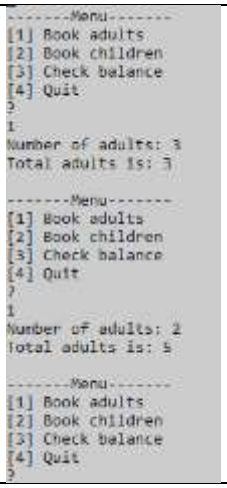
Thereafter the following menu will be displayed (already coded in the program provided):

```
-------Menu-------
[1] Book adults
[2] Book children
[3] Check balance
[4] Quit
?
```

Option 1: The program should

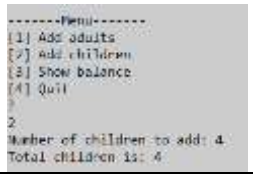
- ask how many adults should be added to the booking;

- read the number from the keyboard;
- increase the total number of adults;
- display the total number of adults.



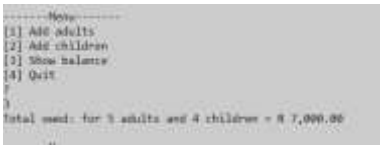
Option 2: The program should

- ask how many children should be added to the booking;
- read the number from the keyboard;
- increase the total number of children;
- display the total number of children.



Option 3: The program should

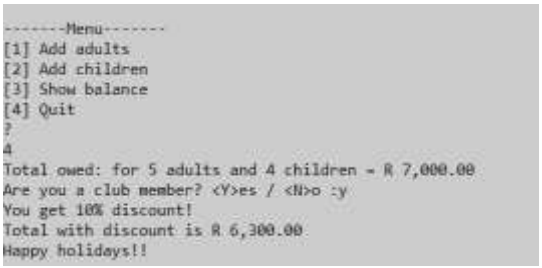
- calculate the total cost for all the adults, all the children for the number of nights they will sleep, and
- display the total cost.



After getting the balance, the user of the program should still be able to add adults and/or children.

Option 4: The program should

- do the same calculation and display of information as for option 3, but the menu should not be displayed again.
- ask if the user is a club member – if the user is a club member, a discount of 10% should be given.



MEMO and mark sheet	mark	student's mark
`import java.text.DecimalFormat;`		
`public class HolidayBooking`		
`{ //sta`		

rt class		
public static void main(String[] args)		
{ //start main method		
Scanner keyboard = new Scanner(System.in);		
DecimalFormat formatter = new DecimalFormat("R #,##0.00");		
//set constant values for cost per night for adults and children		
final double COST_ADULT = 200;		
final double COST_CHILD = 100;		
//declare variable		
int iChoice, iNumAdults, iNumChildren, iTotAdults, iTotalChildren, iNumSleeps;		
double rTotalCost ;		
char cAnswer;		
//initialize variables		
iTotAdul ts = 0 ;	**1**	
iTotalChil dren = 0;	**1**	
rTotalCo st = 0;	**1**	
//read the number of nights a persons wants to sleep on the trip		
System.out.print("How many nights do you want to sleep? :");		
iNumSleeps = keyboard.nextInt();	**1**	
do		
{		
//displ ay menu		
System.out.println("\n-------Menu-------");		
System.out.println("[1] Add adults");		
System.out.println("[2] Add children");		
System.out.println("[3] Show balance");		
System.out.println("[4] Quit\n? ");		
//read menu option		
iChoice = keyboard.nextInt();	**1**	
switch(**iChoice**)	**1**	

{		
case 1:		
{		
//read the number of adults to add to the booking		
System.out.print("Number of adults to add: ");		
iNumAdults = keyboard.nextInt();	**1**	
//increase the number of adults going on the trip		
iTotAdults += iNumAdults;	**1**	
//display the total number of adults going on the trip so far		
System.out.println("Total adults is: " + iTotAdults);		
break;	**0.5**	
}		
case 2:		
{		
//read the number of children to add to the booking		
System.out.print("Number of children to add: ");		
iNumChildren = keyboard.nextInt();	**1**	
//increase the number of children going on the trip		
iTotalChildren += iNumChildren;	**1**	
//display the total number of children going on the trip so far		
System.out.println("Total children is: " + iTotalChildren);		
break;	**0.5**	
}		
case 3:		
case 4 :		
{		
//calculate and display the cost for the trip		
rTotalCost = (iTotAdults * COST_ADULT + iTotalChildren * COST_CHILD) * iNumSleeps;	**2**	
System.out.println("Total owed: for " + iTotAdults + " adults and " +		

iTotalChildren + " children = " + formatter.format(rTotalCost));		
break;		
}		
default: // Invalid choice		
System.out.println("\n-------Invalid choice!\n");		
} //end switch		
}		
while (**iChoice != 4**); //end of the loop displaying the menu options	**1**	
//ask whether the person is a club member?		
System.out.print("Are you a club member? <Y>es / <N>o :");		
cAnswer = Character.toUpperCase(keyboard.next().charAt(0));		
if (cAnswer == 'Y')	**1**	
{		
//calculate discount and display new final amount		
System.out.println("You get 10% discount!");		
rTotalCost *= 0.9;	**1**	
System.out.println("Total with discount is " + formatter.format(rTotalCost));		
}		
System.out.println("Happy holidays!!");		
} //end main method		
} //end class		
	16	0

Question 6 [10]

Match the description in column A to the term in column B.

In the online version of the paper the terms in column B will appear in a random order – different for each student.

	Column A		Column B
A.	Data supplied to a method to perform a specific task.	1.	Argument of a method.
B.	Methods that can be called from a class directly without having to instantiate an object.	2.	Static methods

C.	Methods that perform tasks and do not return any values.	3.	Void methods
D.	Arithmetic operators	4.	+ - / *
E.	Relational operators	5.	< > <= >=
F.	A unary operator	6.	!
G.	A compound operator	7.	*=
H.	A Boolean value containing the result of a Boolean expression	8.	flag
I.	A value that appears directly in an arithmetic expression, for example, the value 20 in the expression (20 + iPrice1 = iPrice2) / 3.0	9.	literal
J.	The type of casting that needs to be done when data from a higher rank is assigned to a variable with data type of a lower rank.	10.	Narrowing casting
		11.	Instance methods
		12.	&&
		13.	Ternary value
		14.	Widening casting
		15.	variable

Question 7 **[1]**

Which of the following Java assignment statements will correctly calculate the result of 120 divided by 7? Assume all variables have been declared with data types as indicated by the naming convention used.

a. `rResult = 120/7.0;`
b. `rResult = 120/7;`
c. `iResult = 120.0/7;`
d. `rResult = 120\7;`

Question 8 **[1]**

When the following formatting pattern is used in a DecimalFormat object: "##.###" how will the value 0.345678 be displayed by the object's format method?

a. 0.346
b. 00.346
c. 00.345
d. 0.346

Question 9 **[2]**

Choose all the equivalent statements to increase the value of rNumber by 1.

a. `rNum ++;`
b. `rNum +=1;`
c. `rNum = rNum + 0.1;`
d. `rNum = rNum * 1.1;`

Question 10 **[2]**

Which two of the following Java code statements will correctly determine the number of hours in a certain number of minutes, as well as the remaining minutes that does not form part of a full hour.

a.
```
iNumHrs = iMin / 60;
iMinRem = iMin % 60;
```
b.
```
iNumHrs = iMin / 60;
iMinRem = iMin – (iNumhrs * 60);
```
c.
```
iNumHrs = iMin \ 60;
iMinRem = iMin MOD 60;
```
d.
```
iNumHrs = iMin / 60;
iMinRem = Math.mod(iMin,60.0);
```

Question 11 [2]

Choose the best Java data type for a variable that will store the number of bags that can be loaded on a truck. The maximum capacity of the truck is 1500 bags.

a. short
b. byte
c. int
d. long

Question 12 [2]

Which two of the following values are equivalent to the value 0.000986?

a. 9.86 * Math.pow(10,-4)
b. 9.86E-4
c. $0.986*10^{-2}$
d. $0.986*10^{-4}$

Question 13 [3]

Which one of the following Java assignment statements correctly represents the mathematical equation?

$$x = 3 + x - \frac{xy}{y+5} \div \sqrt[3]{y}$$

a.
```
3 + x – x * y / (y + 5) / Math.pow(y, 1/3.0);
```
b.
```
3 + x – (x * y) / y + 5 / Math.pow(y, 1/3.0);
```
c.
```
3 + x – (x * y) / (y + 5) / Math.pow(y, 1/3);
```
d.
```
3 + x – x * y / (y + 5) / Math.cube(y);
```

Question 14 [1]

What is the value of the variable y after execution of the following code (enter only the value).

```
double x = 20.5;
double y = ++x;
```

21.5

Question 15 [1]

What is the value of the variable y after execution of the following code (enter only the value).

```
int x = 11;
x *= x;
```

121

Question 16 [2]

What is the value of the variable B after the following Java calculation?

```
double B = 200 * 10 / 2 % 3;
```

1.0

Question 17 [1]

What will be displayed by the following statement?

```
System.out.println((int)Math.floor(89.001));
```

89

Question 18 [3]

What will be displayed by the following statement?

```
System.out.println(Math.round(Math.PI*100)/100.0);
```

3.14

Question 19 [2]

Choose the correct relationship between the two values:

a. "linear" < "liar"
b. "linear" > "liar"
c. "linear" = "liar"
d. "linear" == "liar"

Question 20 [1]

What is the value of the variable x after the following Java calculation?

```
int x = (int) (12/6.0*2);
```

4

Question 21 [2]

Which statement will correctly display the last character in a sentence? Assume all variables have been declared and values read from the keyboard.

a. System.out.println(sSent.charAt(sSent.length()-1));
b. System.out.println(sSent.charAt(sSent.length()));
c. System.out.println(sSent.charAt(sSent.length(-1)));
d. System.out.println(sSent.charAt(sSent.length));

Question 22 [3]

Students writing PPAF05D will pass their module if they meet the following conditions:

- Their final mark is ≥ 50%.
- They scored at least 40% for their final summative mark.

The final mark consists of 60% of the semester work mark + 40% of the final summative mark. Which code segment will correctly and most effectively display a message whether a student passed, or a message why the student did not pass? Assume all calculations and declaration have been done correctly.

a)

```
if (iFinal >= 50)
     if (iFinalSumm >= 40)
          System.out.println("Pass");
     else
          System.out.println("Fail - final summative < 40");
else
     System.out.println("Fail - final mark < 50");
```

b)

```
if (iFinal >= 50 && iFinalSumm >= 40)
     if (iFinalSumm >= 40)
          System.out.println("Pass");
     else
          System.out.println("Fail - final summative < 40");
else
     System.out.println("Fail - final mark < 50");
```

c)

```
if (iFinal >= 50)
     if (iFinalSumm >= 40)
          System.out.println("Pass");
     else
        if (iFinalSumm < 40)
          System.out.println("Fail - final summative < 40");
else
     System.out.println("Fail - final mark < 50");
```

d)

```
if (iFinal >= 50)
     if (iFinalSumm < 40)
          System.out.println("Pass");
     else
          System.out.println("Fail - final summative < 40");
else
     System.out.println("Fail - final mark < 50");
```

Question 23 [2]

Which two Java Boolean expressions correctly determine whether a value falls within the ranges of values represented by the sold lines in the following line graph:



a. (iVal < -2 || iVal >= 2)
b. !(iVal >= -2 && iVal < 2)
c. (iVal < -2 && iVal >= 2)
d. (iVal <= -2 && >= 2)

Question 24 [2]

How many times will the following while loop execute?

```
int iCounter = 2;
while (iCounter >= -5)
{
      System.out.println("The value of iCounter is: " + iCounter);
      iCounter = iCounter - 2;
}
```

4

Question 25 [2]

How many times will the following do-while loop execute?

```
int iCounter = -5;
do
{
      System.out.println("The value of iCounter is: " + iCounter);
      iCounter = iCounter * -2;
}
while (iCounter <= 10);
```

3

Question 26 [2]

How many times will the following for loop execute?

```
int k;
for (k = -1; k <= -10; k ++) System.out.println(k);
```

0

Question 27 [3]

Which set of calculations will correctly calculate the average of the best 5 out of 6 marks? Assume all variables have been declared as indicated by the prefix of the variable names

a)

```
iSum = 0 ;
iSmallest = 101;
for (iCount = 1; iCount <= 6; iCount++)
{
      System.out.print("Enter mark " + iCount + ": ");
      iMark = kb.nextInt();
      iSum += iMark;
      if (iMark < iSmallest) iSmallest = iMark;
}
iSum -= iSmallest;
iAvg = (int) (Math.round(iSum / 5.0));
```

b)

```
for (iCount = 1; iCount <= 6; iCount++)
{
      System.out.print("Enter mark " + iCount + ": ");
      iMark = kb.nextInt();
      iSum += iMark;
      if (iMark < iSmallest) iSmallest = iMark;
```

```
    }
    iSum -= iSmallest;
    iAvg = (int) (Math.round(iSum / 5.0));
```

c)

```
    iSum = 0 ;
    iSmallest = 101;
    for (iCount = 1; iCount <= 6; iCount++)
    {
        System.out.print("Enter mark " + iCount + ": ");
        iMark = kb.nextInt();
        iSum += iSmallest;
        if (iMark < iSmallest) iMark = iSmallest;
    }
    iSum -= iSmallest;
    iAvg = (int) (Math.round(iSum / 5.0));
```

d)

```
    iSum = 0 ;
    iSmallest = 0;
    for (iCount = 1; iCount <= 6; iCount++)
    {
        System.out.print("Enter mark " + iCount + ": ");
        iMark = kb.nextInt();
        iSum += iSmallest;
        if (iMark < iSmallest) iMark = iSmallest;
    }
    iSum -= iSmallest;
    iAvg = iSum / 5;
```

Question 28 [3]

myTUTor instruction – set the marking up to deduct marks for wrong answers. Students select almost all the possible answers.

Review the following problem statement:

Gauteng car Rentals (GCR) rents out small cars at a certain cost per day. In addition to the daily cost, R4.45 is charged for every kilometre covered. An additional cost of 2.6% insurance is charged on the total amount (the daily cost plus the cost per km). A program is needed to calculate the total amount that a customer must pay. Identify all the possible input variables such a program will need:

a. Cost charged per day.
b. The number of kilometres the customer travelled.
c. Number of days the car was rented.
d. Cost charged per kilometre.
e. The insurance percentage
f. Amount owed for all the days a car was rented.
g. Amount owed for all the kilometres travelled.
h. The amount insurance to pay.
i. The total cost for a customer.